

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CE)

Nov 2012

CE305 Computer Networks & Internetworking Layers

Date: 05.11.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following questions.

- (a) List two ways in which the OSI reference model and the TCP/IP reference model are the same. List two ways in which they differ. [04]
- (b) What is the main difference between TCP and UDP? [03]

Q – 2 Answer the following questions.

- (a) What is the remainder obtained by dividing 1101011011 by the generator polynomial $x^4 + x + 1$? And explain the process on the receiving end to calculate CRC. [05]
- (b) Explain various carrier sense multiple access protocols in brief. [05]
- (c) What is the difference between Error correction and error detection technique. List various error correction and error detection techniques. [04]

OR

Q – 2 Answer the following questions.

- (a) Explain simplex stop-and-wait protocol in detail. [05]
- (b) Explain collision-free protocols in detail. [05]
- (c) Explain hamming code in detail. [04]

Q – 3 Answer the following questions.

- (a) Explain IPv4 header in detail. [05]
- (b) Explain the problem of hidden station and exposed station in 802.11 wireless system. [05]
- (c) Explain IPv4 classfull addressing scheme in detail. [04]

OR

Q – 3 Answer the following questions.

- (a) Explain static routing algorithms in detail. [05]
- (b) Compare datagram and virtual-circuit subnets in detail. [05]
- (c) Explain sliding window protocol in detail. [04]

SECTION – II

Q - 4 Answer the following questions.

- (a) Explain broadcast routing in detail. [04]
- (b) Explain flow control at transport layer in brief. [03]

Q - 5 Answer the following questions.

- (a) Explain tunneling at network layer in detail. [05]
- (b) Explain congestion prevention policies at transport, network and data link layer. [05]
- (c) Define autonomous system. Write down the difference between interior gateway protocol and exterior gateway protocol. [04]

OR

Q - 5 Answer the following questions.

- (a) Explain an issue of addressing at transport layer. [05]
- (b) Explain load shedding and jitter control in detail.. [05]
- (c) Explain connection establishment using three-way handshake in TCP. [04]

Q - 6 Answer the following questions.

- (a) Explain HTTP (Hyper Text Transfer Protocol) in detail. [07]
- (b) Explain SMTP (Simple Mail Transfer Protocol) and POP3 in detail. [07]

OR

Q - 6 Answer the following questions.

- (a) Explain resource records of DNS in detail with example. [07]
- (b) Explain hierarchical routing protocol in detail with suitable example. [07]
